Urban Infrastructure Failure Risk — ML Assignment

1. Student Number: 3547228 | ITNPBD6 Assignment

2. Project Methodology

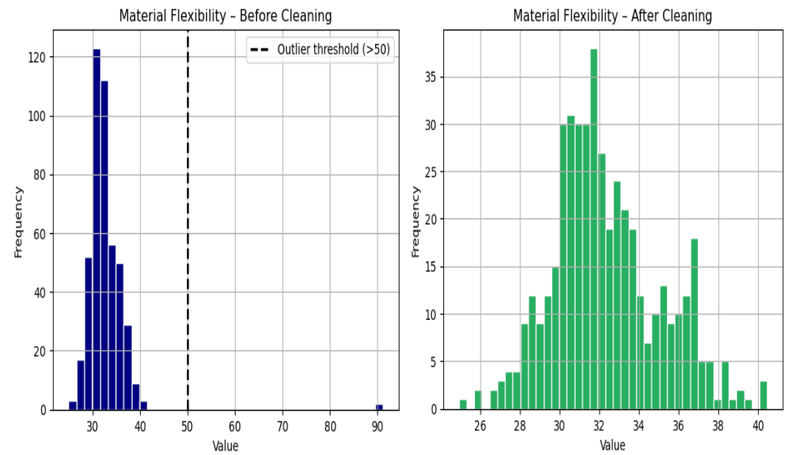
1. **Load & Explore Data:** Read urbandata.csv (456 rows × 12 cols). Check dtypes, class balance (63%/37%), missing values.
2. **Clean Data:** Fix 2 outliers in material_flexibility (>50 → median). Correct typos: 'sumer'→'summer', 'morni7ng'→'morning'. Impute 11 NaN with column medians.
3. **Encode & Scale:** Drop inspector_id (ID, not predictive). One-hot encode 2 categorical columns. Apply StandardScaler for LR & NN.
4. **Split Data:** Stratified 60/20/20 → Train: 273 | Validate: 91 | Test: 92 rows. Test set held out until final evaluation only.
5. **Train & Tune:** Grid search over LR (4 C values), Decision Tree (5 depths), NN (5 configs). Score all on validation AUC-ROC.
6. **Final Evaluation:** Select best model by validation AUC-ROC. Evaluate once on held-out test set.

3. Variables & Type Decisions

Variable	Type
material_flexibility	Numeric
structural_mass_ratio	Numeric
recent_precipitation	Numeric
infrastructure_age	Numeric
surface_debris	Numeric
vibration_exposure	Numeric
asset_density	Numeric
moisture_stress	Numeric
inspection_season	Categorical (OHE)
inspection_window	Categorical (OHE)
inspector_id	DROPPED — ID leakage

Key consequence: One-hot encoding prevents categorical variables from being incorrectly treated as ordinal, while removing inspector_id avoids data leakage and ensures the model generalises to unseen data.

4. Data Preparation — Histogram



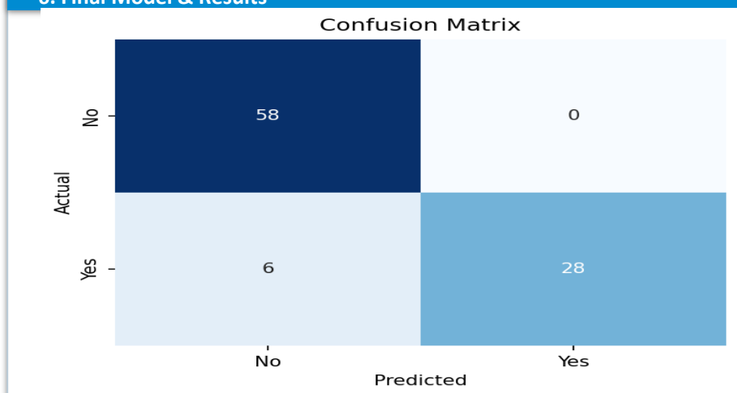
Error detected: 2 values in material_flexibility exceeded 50 (max=91.2), far outside the normal range of 25–45. These were replaced with the column median (31.9) — a standard outlier imputation approach.

5. Model Training & Hyperparameter Tuning

Model	Hyperparameter	Val Accuracy	Val AUC-ROC
Logistic Regression	**C = 0.01**	**92.3%**	**0.978 ★**
Logistic Regression	**C = 0.1**	**92.3%**	**0.978 ★**
Logistic Regression	C = 1.0	91.2%	0.972
Logistic Regression	C = 10.0	90.1%	0.971
Decision Tree	max_depth = 3	82.4%	0.846
Decision Tree	max_depth = 5	81.3%	0.773
Decision Tree	max_depth = 7	80.2%	0.793
Neural Network	layers=(128,64), lr=0.01	92.3%	0.972
Neural Network	layers=(64,32), lr=0.01	87.9%	0.961
Neural Network	layers=(64,32), lr=0.001	84.6%	0.944

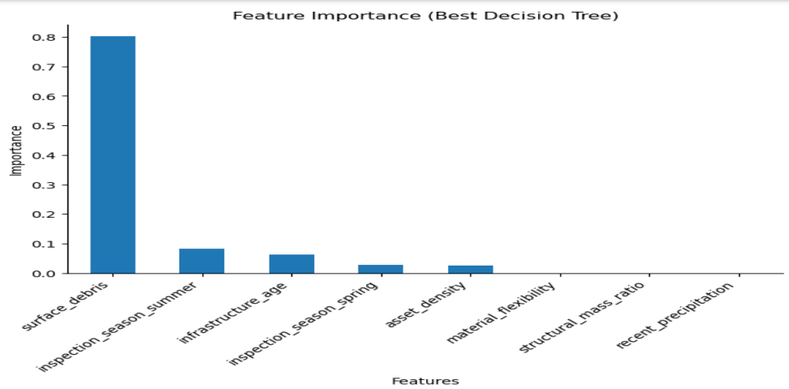
Hyperparameter selection: The hyperparameter values were selected based on commonly recommended ranges in machine learning literature (Géron, 2022). Each model configuration was trained on the 60% training set and evaluated on the 20% validation set. ROC-AUC was used as the evaluation metric because the dataset is moderately imbalanced (63% no-failure / 37% failure), meaning accuracy alone could be misleading. ROC-AUC provides a better measure of how well the model distinguishes between failure and non-failure cases.

6. Final Model & Results



The best-performing model was Logistic Regression (C=0.01), achieving 93.5% test accuracy and an AUC of 0.992. The model produced zero false positives, meaning no unnecessary alerts were triggered. However, six false negatives indicate missed failures, which represent a greater risk in real-world infrastructure systems. Despite this, the high AUC suggests excellent class separation and strong predictive performance

7. Insight from Data & Models



- **surface_debris dominates (80% importance):** This single feature drives ~80% of the Decision Tree's decisions, suggesting debris obstructing drainage and maintenance access is the primary failure trigger. Routine debris clearance could be a highly cost-effective intervention.
- **LR outperforms more complex models:** Despite having far fewer parameters, Logistic Regression (C=0.01) beats both Neural Networks and Decision Trees on AUC-ROC. This implies the decision boundary is approximately linear, strong regularization prevents overfitting on the 273-row training set.
- **Season drives secondary risk:** Summer and spring appear in the top 5 Decision Tree features, suggesting warmer months elevate failure probability, possibly via thermal expansion, increased usage loads, or ground-moisture changes.
- **Operational utility:** With zero false positives and 93.5% test accuracy, the model shows strong potential for deployment; however, further validation on real-world data would be required..

8. References

[1] Géron, A. (2022). Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow (3rd ed.). O'Reilly Media.

[2] Bishop, C. M. (2006). Pattern Recognition and Machine Learning. Springer.

[3] Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. JMLR, 12, 2825–2830.

[4] Fawcett, T. (2006). An introduction to ROC analysis. Pattern Recognition Letters, 27(8), 861–874.